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Configurable modular skate blade assembly with improved damping

Abstract

Disclosed is a configurable modular damping ice-skating blade assembly including a frame with a blade runner extending along a longitudinal axis and being removably mounted in relation to the frame. The modular damping ice-skating blade assembly is configurable with the use of modified blade runners so as to allow one or more configurations for damping the frame in relation to the blade runner.

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Background/Summary

(1) This application is the U.S. national phase of International Application No. PCT/SE2022/050017 filed Jan. 12, 2022, which designated the U.S. and claims priority to SE Patent Application No. 2150034-3 filed Jan. 15, 2021, the entire contents of each of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The invention relates to a configurable modular damping ice-skating blade assembly comprising a frame with a blade runner extending along a longitudinal axis and being removably mounted in relation to the frame.

BACKGROUND

(3) Ice-skating blades, such as for figure skating, ice hockey, speed skating or recreational skating, all evolved from common devices intended to glide on an ice surface for travel and transportation purposes, the first being made from animal bones. Ice-skating blades have ever since been improving in relation with advances in technology and material, and evolving in relation to individual ice sport disciplines. It is obvious that different ice-skating sports and disciplines require different modifications and features to the ice-skating blade. It is preferable to use cushioning equipment and shoe wear for sport activities comprising frequent landing impacts, for instance it is preferable to play basketball in basketball shoes with thick soft outsoles rather than in stiff ballet shoes. While figure skaters can benefit from having ice-skating blades with an integrated damping system, due to on-ice jumping elements, this would be of no benefit for speed skaters who on the other hand benefit from longer blade runners, resulting in increased skating speeds. Common figure skating blades are rigid in construction while some ice hockey blade models and “Klap” speed skating blade models are modified to allow some degree of blade runner movement in relation to the frame or boot, however not primarily intended for damping landing impacts from jumps on the ice.

(4) Conceptually known landing damping ice-skating blades and inline skates enable one configuration of damping that occurs to a lesser extent in a single axis relative to the frame. US patent No. 2017/0028291—a blade arrangement comprising a support for an ice skate boot with a blade runner mounted to the support and a suspension structure arranged between the support and the blade runner.

(5) U.S. Pat. No. 9,089,763—a skate boot appliance for absorbing impact force of landing skating maneuvers by disposing a plunger or displacement member in a receptacle in response to an impact force.

(6) EP patent No. 2,123,334—inline skate having impact absorbing members, made of resilient material, inserted into insert grooves of the frame.

(7) U.S. Pat. No. 9,943,748—a skate blade system wherein the blade portion is fastened at the heel end to a blade housing in a fixed relationship and is unattached from the blade portion blade housing at the toe end.

(8) CA patent No. 2,324,724—ice skate with suspension, accomplished with either coil or leaf type spring, between the skate boot and the blade in a blade runner/holder housing.

(9) U.S. Pat. No. 6,007,075—a skate of the clap type wherein the blade is pivotally movable with respect to the boot.

(10) U.S. Pat. No. 4,993,725—a single blade ice skate formed as a unitary assembly from a flat sheet of metal, wherein the boot plates are bent into a U-shape, so as to provide a U-shaped vertical spring member operating in the vertical plane of the blade.

(11) As figure skating has evolved from being oriented primarily on skating skills, modern figure skaters, soloists and pair skaters, spend significantly more time on practicing on-ice jump elements.

A large number of landing repetitions and absorbing of high ground reaction forces is one of the contributing factors to an increased frequency of chronic overuse injuries in figure skaters over the last decades.

(12) Landings from on-ice figure skating jumps are performed on a single leg backwards with glide away from the place of the impact in a curve. The landing has a toe to heel character and the ground reaction force vector is pointing upwards to different angles, shifting from an acute angle relative to a forward-rearward longitudinal axis of the blade runner at initial contact of the blade runner with the ice surface, to perpendicularly upwards pointing peak force when a skater stabilizes a landing position. The location of the loading on the blade runner is also shifting from the front section of the blade runner to the rear section.

SUMMARY

(13) In accordance with the disclosure, there is provided an assembly having an object of providing an improved and configurable modular ice-skating blade assembly.

(14) In accordance with an embodiment of the invention, this object is obtained by a configurable modular damping ice-skating blade assembly comprising a frame with a blade runner extending along a longitudinal axis and being removably mounted in relation to the frame. The modular damping ice-skating blade assembly is configurable with the use of modified blade runners so as to allow two or more configurations for damping the frame in relation to the blade runner.

(15) In accordance with a further embodiment, the object is obtained by a blade assembly which is further configurable by means of at least one damping unit arranged in said frame, so as to allow one or more configurations for damping the frame in relation to the blade runner, wherein the damping unit is located between protruding arms of the frame, which are shaped to secure the damping unit in the forward-rearward longitudinal axis relative to the frame and, optionally, also to allow lateral displacements of the damping unit when a compressive load is transmitted to the damping unit.

(16) An advantage with the assembly according to the disclosure is that it constitutes a modular ice-skating blade assembly with optional configurations enabling improved impact damping in two degrees of freedom (2DOF), where the blade runner dynamically interacts with the direction and the location of the ground reaction force vector during a landing impact, or in one degree of freedom (1DOF), where the blade runner is pivotally attached in the frame to allow damping of the vertical component of the ground reaction force that occurs in the rear section of the ice-skating blade assembly during a landing impact, wherein the damping unit in the modular ice-skating blade assembly is located between protruding arms of the frame that secure the damping unit in the forward-rearward longitudinal axis relative to the frame while allowing lateral displacements of the damping unit when a compressive load is transmitted to the damping unit during landing impacts and also to accommodate damping units of larger size which would be disadvantageous to encase with the frame, or a configuration where the blade runner and the frame creates a rigid construction with no damping.

(17) According to an embodiment, the assembly may comprise at least one damping unit configured for providing damping of the movement of the frame in relation to the blade runner, and wherein said at least one damping unit is arranged in said frame with a configuration for allowing said blade runner to move in a flexible manner in said frame along the longitudinal axis and along a vertical axis which is perpendicular to said longitudinal axis, so as to provide damping in two degrees of freedom.

(18) According to an embodiment, the assembly may comprise at least one damping unit configured for providing damping of the movement of the frame in relation to the blade runner, and where said blade runner is pivotally coupled in relation to said frame in a configuration to move in a flexible manner in said frame along an axis of rotation, which is transverse to said longitudinal axis, so as to provide damping in one degree of freedom.

(19) According to an embodiment, the blade runner may be rigidly mounted in relation to said

frame creating a rigid configuration with generally no damping.

(20) According to an embodiment, the damping unit may have a curved longitudinal profile that enables the blade runner to be secured within the frame in the forward-rearward longitudinal axis and vertical axis relative to the frame.

(21) According to an embodiment, the frame may comprise a front-frame unit and a back-frame unit and wherein a first damping unit is located between protruding arms in the front-frame unit and a second damping unit is located between protruding arms in the back-frame unit.

(22) According to an embodiment, the contact between said blade runner and said damping unit may be via a supporting part.

(23) According to an embodiment, the damping unit may be provided with a groove on its upper side to secure its position within the frame, said groove being configured to fit in a protrusion in said frame or the heads of fastening members that attaches said frame to the sole of a skate boot.

(24) According to an embodiment, the damping unit may be provided with a protrusion extending from its top portion to secure its position within the frame, said protrusion being configured to fit in an aperture in said frame.

(25) According to an embodiment, the damping unit may be configured as an elastic body with a cavity.

(26) According to an embodiment, the damping unit may comprise an infill or a spring which is accommodated within said cavity in the elastic body, so as to reinforce the damping unit.

(27) According to an embodiment, the damping unit and said infill may be made of elastomeric material of different A Shore hardness.

(28) According to an embodiment, the blade runner may be provided with at least one large transversally extending aperture allowing the blade runner to be attached to said frame, or said blade runner may be provided with at least one large transversally extending aperture and one generally circular through-hole allowing the blade runner to be attached to said frame, said through-hole being intended for the pivotal coupling with the frame, or said blade runner may be provided with at least two generally circular through-holes allowing the blade runner to be attached to said frame creating a rigid construction.

(29) According to an embodiment, the blade runner may be attached to the frame by means of a transversal bearer extending through said large aperture or said circular through-hole.

(30) According to an embodiment, the large transversally extending aperture may allow movement of the blade runner upwards to different angles in the forward-rearward longitudinal axis, the vertical axis and along an axis of rotation relative to the frame without interfering with the transversal bearer that passes through the large aperture.

(31) According to an embodiment, the transversal bearer may be configured as a pin with a screw hole and a screw located in said screw hole, so as to create a counterforce mechanism enabling to adjust the fit and preloading force between the blade runner and the damping unit by means of screwing said screw from said pin against said blade runner which shifts said blade runner upwards in the direction of said damping unit.

(32) According to an embodiment, a friction part or friction points may be used and configured to reduce the friction and to eliminate direct contact between the blade runner and the inside walls of the frame.

(33) According to an embodiment, the frame may comprise bumpers extending downwards from a top inside wall of said frame and in line with the blade runner, and being configured for protecting the frame from possible impacts of the blade runner during its movement inside the frame.

(34) According to an embodiment, the frame may constitute an integral part of a sole of an ice-skating boot.

(35) According to an embodiment, damping units of different stiffnesses may be arranged for skaters of different weight categories, generally limited by the displacement of the damping unit being less than one millimeter under a load of two times the body weight of a skater.

- (36) According to an embodiment, the protruding arms of the frame may be joined to form an opening which can accommodate the damping unit.
- (37) According to an embodiment, the damping unit may be located between protruding arms of the frame, which are shaped to allow lateral displacements of the damping unit when a compressive load is transmitted to the damping unit.
- (38) According to an embodiment, the ice-skating blade assembly may comprise a frame with a blade runner extending along a longitudinal axis, wherein the frame comprises one or more protruding arms which are shaped to secure a damping unit in the forward-rearward longitudinal axis relative to the frame.
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Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is an elevational view of a modular damping ice-skating blade assembly in configuration for improved impact damping in two degrees of freedom in use on a skate boot according to a first embodiment of the present invention;
- (2) FIG. 2 is an exploded perspective view of the configurable modular damping ice-skating blade assembly of FIG. 1 in configuration for improved impact damping in two degrees of freedom according to the first embodiment of the present invention;
- (3) FIG. 3 is a further exploded perspective view of the configurable modular damping ice-skating blade assembly of FIG. 1 in configuration for improved impact damping in two degrees of freedom according to the first embodiment of the present invention, as regarded from slightly different angles as compared with the figures mentioned above;
- (4) FIG. 4 is a further exploded perspective view of the configurable modular damping ice-skating blade assembly of FIG. 1 in configuration for improved impact damping in two degrees of freedom according to the first embodiment of the present invention, as regarded from slightly different angles as compared with the figures mentioned above;
- (5) FIG. 5A is an elevational view of a blade runner shown in FIG. 1 configured for improved impact damping in two degrees of freedom according to the first embodiment of the present invention;
- (6) FIG. 5B is an elevational view of an alternative configuration of the blade runner shown in FIG. 5A configured for improved impact damping in one degree of freedom according to the first embodiment of the present invention;
- (7) FIG. 5C is an elevational view of an alternative configuration of the blade runner shown in FIG. 5A configured for rigid construction with no damping according to the first embodiment of the present invention;
- (8) FIG. 6 is a cross-sectional view of the modular damping ice-skating blade assembly of FIG. 1 in configuration for improved impact damping in two degrees of freedom according to the first embodiment of the present invention;
- (9) FIG. 7 is a cross-sectional view of an alternative configuration of the configurable modular damping ice-skating blade assembly of FIG. 1 in configuration for improved impact damping in one degree of freedom according to the first embodiment of the present invention;
- (10) FIG. 8 is a cross-sectional view of an alternative configuration of the configurable modular damping ice-skating blade assembly of FIG. 1 in configuration for rigid construction with no damping according to the first embodiment of the present invention;
- (11) FIG. 9 is an elevational view of a further embodiment of the configurable modular damping ice-skating blade assembly shown in FIG. 1 in configuration for improved impact damping in two degrees of freedom;
- (12) FIG. 10 is an exploded perspective view of the configurable modular damping ice-skating

blade assembly shown in FIG. 9 in configuration for improved impact damping in two degrees of freedom according to the further embodiment of the present invention;

(13) FIG. 11 is an exploded perspective view of an alternative configuration of the configurable modular damping ice-skating blade assembly shown in FIG. 9 in configuration for improved impact damping in one degree of freedom according to the further embodiment of the present invention;

(14) FIG. 12 is a cross-sectional view of the configurable modular damping ice-skating blade assembly shown in FIG. 11 in configuration for improved impact damping in one degree of freedom according to the further embodiment of the present invention;

(15) FIG. 13A is a perspective view of a damping unit shown in FIG. 1 according to the first embodiment of the present invention;

(16) FIG. 13B is a perspective view of an alternative configuration of the damping unit shown in FIG. 13A according to the first embodiment of the present invention;

(17) FIG. 13C is an exploded perspective view of an alternative configuration of the damping unit shown in FIG. 13A according to the first embodiment of the present invention;

(18) FIG. 13D is a perspective view of a further embodiment of a damping unit shown in FIG. 13A;

(19) FIG. 13E is a perspective view of a further embodiment of a damping unit shown in FIG. 13A;

(20) FIG. 13F is an elevational view of an alternative embodiment of a damping unit shown in FIG. 13A;

(21) FIG. 13G is a front view of the embodiment of the damping unit shown in FIG. 13F;

(22) FIG. 13H is a perspective view of the embodiment of the damping unit shown in FIG. 13F;

(23) FIG. 13I is a perspective view of an alternative embodiment of the damping unit shown in FIG. 13F;

(24) FIG. 13J is an exploded perspective view of the embodiment of the damping unit shown in FIG. 13I;

(25) FIG. 13K is a further exploded perspective view of the embodiment of the damping unit shown in FIG. 13I, as regarded from different angle as compared with the FIG. mentioned above;

(26) FIG. 13L is a perspective view of an alternative embodiment of the damping unit shown in FIG. 13I;

(27) FIG. 13M is an exploded perspective view of the embodiment of the damping unit shown in FIG. 13L;

(28) FIG. 13N is a further exploded perspective view of the embodiment of the damping unit shown in FIG. 13L, as regarded from different angle as compared with the figure mentioned above;

(29) FIG. 13O is a perspective view of an alternative embodiment of the damping unit shown in FIG. 13L;

(30) FIG. 13P is an exploded perspective view of the embodiment of the damping unit shown in FIG. 13O;

(31) FIG. 13Q is a further exploded perspective view of the embodiment of the damping unit shown in FIG. 13O, as regarded from different angle as compared with the figure mentioned above;

(32) FIG. 14A is a perspective view of an alternative embodiment of the pin shown in FIG. 10 in configuration with a screw;

(33) FIG. 14B is an exploded perspective view of the pin with the screw shown in FIG. 14A;

(34) FIG. 14C is a cross-sectional view of the pin with the screw shown in FIG. 14A;

DETAILED DESCRIPTION

(35) According to an embodiment, the invention provides a configurable modular damping ice-skating blade assembly for use with modified blade runners for optional configurations enabling a unique impact damping system in two degrees of freedom (2DOF), where a blade runner dynamically interacts with the direction and the location of the ground reaction force vector (GRFV) during a landing impact by moving in a frame upwards to different angles in a forward-

rearward longitudinal axis and a vertical axis which is substantially perpendicular to the frame, against flexible damping units placed in the frame of the blade assembly, or in one degree of freedom (1DOF), where a modified blade runner is pivotally attached in the frame about an axis transverse to the forward-rearward longitudinal axis of the blade runner, against flexible damping units placed in the frame, to allow damping of the vertical component of the ground reaction force that occurs in the rear section of the blade assembly during a landing impact, but also a configuration where a modified blade runner and the frame of the blade assembly creates a rigid construction with no damping.

(36) The present invention is described in the following examples, which may represent more than one embodiment of the present invention.

(37) Referring to the figures above, the present invention relates to a configurable modular damping ice-skating blade assembly **90** for use on a skate boot **100**, which comprises a frame **40**, part that is attached to the sole of the skate boot **100**, a blade runner **70**, part for gliding on an ice surface, and flexible damping units **50** to reduce landing impact loads, which are placed in the frame **40**, where the modular damping ice-skating blade assembly **90** is configurable with the use of modified blade runners **70-1**, **70-2** or **70-3** for damping in 2DOF and 1DOF, or a rigid construction with no damping. The blade runners **70-1**, **70-2** or **70-3** are removably mounted to the frame **40**, however the blade runners **70-1** or **70-2** are not rigidly attached to it, allowing slidable movement inside the frame **40**, as shown in FIGS. **6-8**. During a landing impact from a jump, the GRFV at initial contact of the blade runner **70** with an ice surface points upwards at an acute angle relative to the frame **40**, along a reference axis (F), and perpendicularly upwards, along a reference axis (Y), when a skater stabilizes a landing position. The 2DOF damping configuration of the modular damping ice-skating blade assembly **90** includes the blade runner **70-1** with two large apertures **73** to allow for slidable movement of the blade runner **70-1** in the forward-rearward longitudinal axis (X) and the vertical axis (Y) inside the frame **40** against the damping units **50**, as shown in FIGS. **1-4**, **6**, **9** and **10**. The 1DOF damping configuration of the modular damping ice-skating blade assembly **90** includes the blade runner **70-2** with one large aperture **73** and one through-hole **78**, generally of circular cut-out located in the front section of the blade runner **70-2**, intended for pivotal coupling **80** with the frame **40**, enabling pivotal slidable movement of the blade runner **70-2** in the axis of rotation (P), which is transverse to the longitudinal axis of the blade runner **70-2**, inside the frame **40** against the damping units **50**, generally one damping unit **50** placed in the rear section of the frame **40**, as shown in FIGS. **7**, **11** and **12**. The configuration enabling the rigid construction of the modular damping ice-skating blade assembly **90** includes a blade runner **70-3** with two through-holes **78**, generally of circular cut-out which does not permit an upward movement of the blade runner **70-3** inside the frame **40**, generally one located in the front section and the other in the rear section of the blade runner **70-3**, where such configuration of the modular damping ice-skating blade assembly **90** may not contain damping units **50**, as shown in FIG. **8**. In an alternative embodiment, the blade assembly **90** is not configurable and is set up for only one configuration, either for damping in 2DOF or 1DOF or a rigid construction without damping.

(38) The damping units **50**, that are placed in the frame **40** in the 2DOF and 1DOF damping configurations, support at least a portion of the top side of the blade runner **70-1** or **70-2**, and secure the blade runner **70-1** or **70-2** in the forward-rearward longitudinal axis (X) and the vertical axis (Y) relative to the frame **40**. In an alternative embodiment, the contact between the blade runner **70-1** or **70-2** and the damping unit **50** may be via a supporting part **63**, as shown in FIG. **9-12**. The frame **40** prevents the blade runner **70** from longitudinal bending, twisting, rotating and moving side-to-side in transversal axes relative to the frame **40** by covering at least a portion of the blade runner **70** from sides in transversal axes relative to the blade runner **70**. The contact between the inside walls **26** of the frame **40** and the blade runner **70** is rather tight to prevent wobbling of the blade runner **70** inside the frame **40**, but loose enough to allow for sliding movements of the blade

runner **70-1** or **70-2** under high loads during landing impacts in the 2DOF and 1DOF damping configurations.

(39) The blade runner **70** is placed inside the frame **40** and secured by a transversal bearer **35** that passes through the large aperture **73** or through-hole **78** in the blade runner **70**. The large apertures **73** in the blade runner **70-1** and **70-2** must be of geometry and dimensions that will allow movement of the blade runner **70-1** and **70-2** inside the frame **40** without interfering with the transversal bearer **35**, upwards to different angles in the forward-rearward longitudinal axis (X) and the vertical axis (Y) relative to the frame **40** in the 2DOF damping configuration using the blade runner **70-1**, or pivotally upwards in the axis of rotation (P) relative to the pivotal coupling **80** generally located in the front section of the frame **40** in the 1DOF damping configuration using the blade runner **70-2**, best shown in FIGS. **6** and **7**. According to an embodiment, a gap must be left between the upper side of the blade runner **70** and the top inside surface of the frame **40** to allow for the upward movement of the blade runner **70-1** or **70-2** inside the frame **40**, without interfering with the top inside surface of the frame **40**. In the configuration where the blade runner **70-3** and the frame **40** creates a rigid construction, the transversal bearers **35** that passes through the through-holes **78** in the blade runner **70-3** are in tight contact with the through-holes **78** which does not permit any movement of the blade runner **70-3**, best shown in FIG. **8**.

(40) The frame **40** comprise one or more protruding arms **32** and **33**, directed downwardly from mounting plates **22** towards the blade runner **70**, which are advantageously shaped to install and secure the damping unit **50** in the forward-rearward longitudinal axis (X) relative to the frame **40** against excessive movements in the assembly **90** or collisions with the blade runner **70** located on the skater's second leg, which may occur during skating movements, while allowing lateral displacements of the damping unit **50**, best shown in FIGS. **1**, **2** and **9-12**. When the damping unit **50** is inserted into the frame **40**, it is supported by the frame **40** from its upper side. However, the frame **40** of the modular damping ice-skating blade assembly **90** is not encasing the damping unit **50** laterally from the sides in transversal axis relative to the damping unit **50** and a gap is left between the front and rear portions of the damping unit **50** and the protruding arms **32** and **33**. This way, the frame **40** does not interfere with lateral displacements of the damping unit **50** which occur due to its flexible nature, when a compressive load is being transmitted from the blade runner **70-1** or **70-2** to the damping unit **50** during landing impacts, and also accommodates damping unit **50** of larger sizes which would be disadvantageous to encase with the frame **40** as it would increase the weight of the assembly. It should now be clear that the phrase “install and secure” is used to define that the damping unit **50** is accommodated between the protruding arms **32**, **33** in a manner in which it may or may not be in contact with either one, or both, of said protruding arms **32**, **33**. In an alternative embodiment, the damping unit **50** may be partially covered laterally from sides or be in tight contact with the protruding arms **32** and **33**, such as in the case of securing the damping unit **50** within the frame **40**, without dramatically affecting the lateral displacements of the damping unit **50** during compressive loading. In an alternative embodiment, the damping unit **50** may extend into a slot **18** in the protruding arms **32** and **33** to further secure its position within the frame or to provide further damping. The protruding arms **32** and **33** may be joined to form an opening **25** shaped to conform to the longitudinal profile of the damping unit **50**, best show in FIG. **2**. The greater the load the greater the displacement of the damping unit **50**. In the 2DOF damping configuration, the load can be equally distributed by the blade runner **70-1** to all damping units **50** in the frame **40**, such as during upright stands on the ice, or any of the damping units **50** placed in the frame **40** can be exposed to a larger portion of the load than the other, such as during spins or take-offs where the load is generally situated in the front part of the blade runner **70**.

(41) —Frame—

(42) As presented in FIG. **1**, the frame **40** generally consists of mounting plates **22** for attaching to the forward and heel portions of the sole of the skate boot **100** by using various fastening members, such as but not limited to screws or rivets. The mounting plates **22**, typically planar surfaces,

include a plurality of holes **9** for fastening members, best shown in FIGS. 2-4. In an alternative embodiment, the frame **40** may be an integral part of the sole of the skate boot **100**, lowering the weight of the complex by excluding the mounting plates **22** and the need for fastening members for attaching the frame **40** to the sole of the skate boot **100**.

(43) The frame **40** of the modular damping ice-skating blade assembly **90**, in this example as shown in the drawings, is divided into a front-frame **20** spaced from back-frame **60**. In an alternative embodiment, the frame **40** may be composed from a different number of separated frames or composed of a single-piece frame. The front-frame **20** and back-frame **60** may include side frames, front-side-frame **21** and back-side-frame **61**, according to an embodiment best shown in FIGS. 1-4. In an alternative embodiment, the frame **40** may include a different number of side frames. The front-side-frame **21** and back-side-frame **61** are connected to the respective front-frame **20** and back-frame **60** by one or more fastening members, in this example screws **30** each passing through one frame hole **31** of the side frames **21** and **61**. The frames **20** and **60** in connection with the side frames **21** and **61** cover at least a portion of the blade runner **70** from both sides in transversal axis relative to the blade runner **70**, to prevent the blade runner **70** from longitudinal bending, twisting, rotating and moving side-to-side in transversal axes relative to the frame **40**. In an alternative embodiment, the frame **40** does not include a side frame, as shown in FIGS. 9-11, and at least a portion of the blade runner **70** is secured in the slot **18** in the protruding arms **32** and **33** from both sides in transversal axes relative to the blade runner **70**, best shown in FIGS. 10 and 11.

(44) The frame **40** is formed of a stiff, lightweight material such as, but not limited to, aluminum, titanium, polymer material or composite materials. The frame **40**, as presented in FIGS. 1-4 and 6-8, is 3D printed from Nylon **12**. The construction of the frame **40**, being 3D printed from Nylon **12**, is able to withstand the high loads that occur on the ice during skating, spinning and jumping. The frame **40**, as presented in FIGS. 9-12, is CNC machined from aluminum. The entire assembly **90**, in this example, comprising the frame **40**, the damping units **50** and the blade runner **70** weighs less than a conventional figure skating blade (not shown) of corresponding size made of stainless steel or carbon steel.

(45) The transversal bearer **35** that holds the blade runner **70** within the frame **40** may be in the form of various fastening members, generally of cylindrical shape made of stiff material such as, although it is not limited to, aluminum, titanium, stainless steel, carbon steel or composite materials. The transversal bearer **35** may be an integral part of the frame **40**. According to the first embodiment of the invention, the transversal bearer **35** that holds the blade runner **70** within the frame **40** has a form of a projecting portion **23** of cylindrical shape extending from the inside wall **26** of the frames **20** and **60**, located in the front bottom sections of the frames **20** and **60**, best shown in FIGS. 2-4. In an alternative embodiment, the frame **40** may include a different number of projecting portions **23**, in different sections of the frame **40**. The projecting portions **23** are extending to a distance enabling to mount the blade runner **70** to the frame **40**, in this example 4 mm. The projecting portions **23** may include a thread **24** for attaching the screws **30** passing through the frame hole **31** of the side frames **21** and **61**, as shown in FIG. 2-4. The projecting portions **23** may extend to fit into the side frames **21** and **61**. In an alternative embodiment shown in FIGS. 6 and 7, the transversal bearer **35** may be a pin **36** with a groove **38** that can pass through the frame hole **31** located in one or both of the protruding arms **32** and **33** of the front and back-frame **20** and **60**, and the large apertures **73** or through-holes **78** of the blade runner **70** securing it within the frame **40**. Once the pin **36** is installed, it is secured on the other side of the frame **40** by a retaining ring **39** that fits into the groove **38** of the pin **36**, preventing it from being pulled out from the frame **40**. In an alternative embodiment shown in FIGS. 14A-14C, the transversal bearer **35**, such as the pin **36** in this example, is constructed with a screw hole **37**, passing vertically along the (Y) axis through the centre of the pin **36**, adapted to receive a screw **34**. The screw **34** is of a length that fits completely inside the pin **36**, so that the pin **36** with the screw **34** can easily pass through

the frame hole **31** for installation in the frame **40**. The pin **36** with the screw **34** is of use in the 2DOF and 1DOF damping configurations of the modular damping ice-skating blade assembly **90**, when using the blade runner **70-1** or **70-2** with the large apertures **73**. While the pin **36** remains in fixed position relative to the frame **40**, the screw **34** can be screwed upwards in vertical axis (Y) against the blade runner **70-1** or **70-2** that sits with the top inside surface of the large aperture **73** on the top of the pin **36** when the blade runner **70-1** or **70-2** is installed to the frame **40**. When the screw **34** is in use and screwed upwards then the blade runner **70-1** or **70-2** is no longer in contact with the top of the pin **36**, but instead with the top of the screw **34**. The pin **36** with the screw **34** thus creates a mechanism enabling to adjust the fit between blade runner **70-1** or **70-2** and the damping unit **50** as well as adjusting the preloading force on the damping unit **50** by shifting the blade runner **70-1** or **70-2** upwards in the direction of the damping unit **50**. The position of the screw **34** in the pin **36** can be adjusted by accessing the hex or slot recess **42** in the lower part of the screw **34** with a fitting key (not shown) via the large aperture **73** in the blade runner **70-1** or **70-2**.

(46) According to the first embodiment of the invention, the inside walls **26** of the frame **40** may be equipped with friction points **27** extending from the inside walls **26** up to several millimeters to reduce the contact area and friction between the blade runner **70** and the frame **40** upon the sliding movement of the blade runner **70** inside the frame **40** during the damping, best shown in FIGS. 2-4. The friction points **27**, in this example, are an integral part of the frame **40**, extending 0.5 millimeters from the inside walls **26** of the frame **40**. In an alternative embodiment, a friction part **67** made of material with low coefficient of friction is used to eliminate direct contact between the blade runner **70** and the frame **40** as well as to reduce the contact area and friction, as shown in FIGS. 11 and 12. In an alternative embodiment, the supporting part **63** placed between the blade runner **70** and the damping unit **50**, preferably made of material with low coefficient of friction, in this case made of Nylon **12**, may also serve to reduce the friction during the sliding movement of the blade runner **70** inside the frame **40**, shown in FIGS. 9-12. In this example, the supporting part **63** and the friction part **67** both comprise a slot **64** to sit tight on the top portion of the blade runner **70**, covering the blade runner **70** from sides in transversal axes relative to the blade runner **70**, in the contact areas with the inside walls **26** of the frame **40**. The outside walls **65** of the supporting part **63** and the friction part **67** are in tight contact with the inside walls **26** of the frame **40**, however sliding movements under high loads during landing impacts are made possible.

(47) In the 1DOF damping configuration of the modular damping ice-skating blade assembly **90**, a bushing **68** may be included in the through-hole **78** for the pivotal coupling **80**, to lower the friction and wear between the blade runner **70-2** and the transversal bearer **35**. The bushing **68** may be an integral part of the friction part **67**, that projects from either side of the inside walls of the slot **64** into the through-hole **78** of the blade runner **70-2**, as shown in FIGS. 11 and 12.

(48) —Blade Runner—

(49) The blade runner **70** is generally an elongated part whose bottom portion is in contact with an ice surface during skating. The blade runner **70** is typically made of, although it is not limited to, stainless steel or high carbon steel. The front portion of the blade runner **70** has an angled front surface which has a plurality of teeth **76** projecting from it. The teeth **76** are the first portion of the blade runner **70** to contact an ice surface during a landing impact from a jump on the ice. The blade runner **70** modified for damping in 2DOF **70-1**, 1DOF **70-2** and for rigid construction **70-3** is best shown in FIGS. 5A-5C. The upper portion of the blade runner **70** where it is to be in contact with the damping unit **50**, possibly via the supporting part **63**, has advantageously shaped profile **75** matching the longitudinal profile of the damping unit **50**, in this case the blade runners **70-1** or **70-2**. The contact between the blade runner **70** and the damping unit **50** does not have to be along the entire length of the damping unit **50**. The blade runner **70** may include one or more cut-out pieces **77** intended to lower the weight of the blade runner **70**.

(50) In this example, depending on the configuration, the blade runners **70-1** and **70-2** are constructed with large apertures **73**, through which passes the transversal bearer **35** for mounting in

the frame **40**, and which further enable a slidable movement of the blade runner **70-1** and **70-2** inside the frame **40**. In an alternative embodiment, the blade runner **70** is constructed with hooks instead of the large apertures **73**, extending from the top portion of the blade runner **70** enabling mounting to the transversal bearer **35** by hooking to it. In an alternative embodiment, the blade runner **70** is constructed with a combination of hook and the large aperture **73**. It can be seen that the position of the large apertures **73** or the through-holes **78** in the blade runner **70** corresponds with the intended location of the transversal bearer **35** in the frame **40**. When the blade runner **70** is installed to the frame **40**, the top surface of the transversal bearer **35** and the top inside surface of the large aperture **73** in the blade runner **70-1** and **70-2** are aligned and in contact. In this example, the apertures **73** in the blade runner **70-1** and **70-2** are of a triangular shape. The large apertures **73** in the blade runner **70-1** and **70-2**, in this example, allow the blade runner **70-1** and **70-2** to move upwards up to 7 millimeters inside the frame **40**. In the 2DOF damping configuration, the large apertures **73** in the blade runner **70-1** enable movement upwards in a range of 104° degrees angle in the forward-rearward longitudinal axis (X) relative to frame **40** without interfering with the transversal bearer **35**. In the 2DOF damping configuration, the connection places of the blade runner **70-1** and the transversal bearer **35** in the frame **40** may also act as pivoting points to further improve the damping interaction between the blade runner **70-1** and the location of the loading on the blade runner **70-1** during a landing impact.

(51) —Damping Unit—

(52) In the 1DOF damping configuration, according to an embodiment of the invention, generally one damping unit **50** is placed in the rear section of the frame **40**, supporting at least a portion of the top side of the blade runner **70-2**, possibly via a supporting part **63**. In an alternative embodiment, two damping units **50** may be used in the 1DOF damping configuration, same as is presented in the 2DOF damping configuration. In the 2DOF damping configuration, according to an embodiment of the invention, generally one damping unit **50** is placed in the front section of the frame **40** and one in the rear section of the frame **40**, together supporting at least a portion of the top side of the blade runner **70-1**, possibly via the supporting part **63**. The damping units **50**, placed in the front and the rear section of the frame **40**, may be of different longitudinal length, best shown in FIGS. **9**, **10**, **13E** and **13D**. The front section of the frame **40** generally accommodates a more elongated damping unit **50** compared to the rear section of the frame **40**. In general, different frame **40** sizes will accommodate damping units **50** of different sizes and lengths. A damping unit **50** of larger size will accommodate a greater range of damping motion for the blade runner **70-1** or **70-2**. If more damping units **50** are placed in the frame **40**, they may be connected to each other by any kind of linkage. According to a further embodiment, one single damping unit **50** can be used for 2DOF damping configuration. In such case the damping unit **50** would be centered in the modular damping ice-skating blade assembly **90**. In such a configuration, the damping unit would suitably be elongated all the way from the front aperture **73** or through-hole **78** to the rear aperture **73** of the blade runner **70-1** or **70-2**, i.e., taking into account the position of the apertures **73** and through-hole **78** in the enclosed drawings.

(53) The lower portion of the damping unit **50** generally has a curved longitudinal profile that enables to secure the blade runner **70-1** or **70-2** within the frame **40** in the forward-rearward longitudinal axis (X) and vertical axis (Y) relative to the frame **40**, as shown in FIGS. **13A-13Q**. In an alternative embodiment, the damping unit **50** consists of two straight portions angled relative to each other, securing the blade runner **70-1** or **70-2** in the forward-rearward longitudinal axis (X) and vertical axis (Y) relative to the frame **40**. In an alternative embodiment, the damping unit **50** may consist of separated parts, one securing the blade runner **70-1** or **70-2** in the forward-rearward longitudinal axis (X) and the other part in the vertical axis (Y) relative to the frame **40**.

(54) The damping unit **50** can be secured within the frame **40** by various methods, such as but not limited to, a frame **40** covering the damping unit **50** laterally from the sides, a fastening member securing the damping unit **50** within the frame **40**, a damping unit **50** glued to the frame **40** or a

damping unit **50** having a rounded top side that fits into a corresponding rounded top inside of the frame **40**. In an example shown in FIGS. **13A-13D**, a groove **53-1** or **53-2** is located on the top side of the damping unit **50**. The groove **53-1** is configured to fit in a protrusion **28** extending downwards from the top inside surface of the frames **20** and **60** of the first embodiment, best shown in FIG. **4**. The damping unit **50** having a groove **53-2** can secure its position relative to the heads of fastening members (not shown) traveling through holes **9** in the mounting plates **22** for attaching the frame **40** to the sole of the skate boot **100**. The damping unit **50** may also comprise protrusions **52** extending from its top portion, to secure the position of the damping unit **50** relative to an aperture **19** in the mounting plate **22** of the frame **40**, according to an embodiment best shown in FIG. **10**. The protrusions **52** and the aperture **19** can be of various shapes, such as but not limited to the protrusions **52** shown in FIGS. **13F-13J**, **13L** and **13M**. The damping unit **50** may also secure its position relative to the blade runner **70**. The damping unit **50**, according to an embodiment best shown in FIGS. **2, 3, 13A-13C, 13G, 13K** and **13N**, covers a small portion of the blade runner **70** from sides in transversal axis relative to the blade runner **70**, by projecting portions **51** extending downwards from the front bottom surface of the damping unit **50**. The projecting portions **51** of the damping unit **50** are spaced in transversal axis according to the thickness of the blade runner **70**. In an alternative embodiment, the blade runner **70-1** or **70-2** may fit by at least a portion into the damping unit **50**, or the damping unit **50** may fit by at least a portion into the blade runner **70-1** or **70-2**. In an alternative embodiment, the supporting part **63** placed between the blade runner **70-1** or **70-2** and the damping unit **50** secures the position of the parts, where the upper portion of the supporting part **63**, where it is to be in contact with the damping unit **50**, has a matching shape **66** to the longitudinal profile as well as to the lower portion of the damping unit **50**, as shown in FIG. **9-11**. The supporting part **63** may affect the stiffness of the damping, for example by increasing the contact area with the damping unit **50**, or by allowing for use of a wider damping unit **50**.

(55) The damping unit **50** used in the modular damping ice-skating blade assembly **90** may consist of one part of one material, or of several parts and materials, wherein the damping unit **50** reacts flexibly to the load caused by the effects of the landing impact. The flexible response of the damping unit **50** to the compressive load transmitted from the blade runner **70-1** or **70-2** during landing impacts, such as displacement and compression, depends on the material properties and the construction of the damping unit **50**. According to an embodiment, the damping unit **50** consists of an elastic body **54** made of an elastomeric material. According to a further embodiment, the elastic body **54** may be an air-filled inflatable bag. The elastic body **54** can be of various shapes, such as but not limited to a cylindrical shape and a curved longitudinal profile, shown in FIGS. **1-4, 6, 7, 9-12** and **13A-13E**. According to an embodiment, the elastic body **54** may be of a ring shape, shown in FIGS. **13F-13K**, or an open oval ring shape, shown in FIGS. **13L-13N**, or of a recurved bow shape, shown in FIGS. **13O-13Q**. According to an embodiment best shown in FIGS. **13A** and **13D**, the elastic body **54** of the damping unit **50** includes a cavity **55** extending longitudinally along the entire length of the elastic body **54**. In an alternative embodiment, the cavity **55** in the elastic body **54** may be extending transversally to the longitudinal axis (X), as shown in FIGS. **13J, 13M** and **13P**. The cavity **55** in the elastic body **54** is intended to affect the stiffness and the flexible response of the damping unit **50** to the compressive load. The cavity **55** may remain empty to reduce the stiffness of the damping unit **50**, or a reinforcing part, such as an infill **56-1** or a spring **56-2**, may be inserted into the cavity **55** to increase the stiffness of the damping unit **50**. The cavity **55** may be of various shapes and sizes and may not have to extend along the entire length or thickness of the elastic body **54**. In an alternative embodiment, several cavities may be included in the elastic body **54**. According to an embodiment shown in FIG. **13C**, the damping unit **50** consists of two separable parts, the elastic body **54** and the infill **56-1** placed inside the cavity **55**. In this example, the elastic body **54** and the infill **56-1** are made of thermoplastic polyurethane (TPU) of different A Shore hardness. According to an embodiment shown in FIGS. **13I-13Q**, the damping unit **50** consists of

three separable parts, the elastic body **54**, the spring **56-2** placed inside the cavity **55** of the elastic body **54** and a side part of the elastic body **54-1** enclosing the spring **56-2** inside the elastic body **54**. The side part of the elastic body **54-1** is configured to fit into the elastic body **54** and may be adhered to it. In these examples, the elastic body **54** and the side part of the elastic body **54-1** are made of elastomeric material such as, but not limited to, TPU or rubber, and the spring **56-2** is formed of a stiff elastic material such as, but not limited to, spring steel, polymer materials or composite materials. The spring **56-2** can be of a leaf spring design or a coil spring design and in various combination with the elastic body **54** of elastomeric material. In an alternative embodiment, the spring **56-2** may be used without the elastic body **54**.

(56) The damping unit **50** is subjected to great compressive loading repeatedly in a cold humid environment of ice rinks. The landing impacts from figure skating jumps can be greater than five times the body weight of a skater, all being transferred to the damping units **50** of just one modular damping ice-skating blade assembly **90** configured for 2DOF or 1DOF damping. The elastomeric material suitable for this purpose is of Shore A hardness **60-100**, with high impact resilience, excellent cold resistance and low compression set, such as, but not limited to, polycyclopentene rubber (CPR) and versions of thermoplastic polyurethane (TPU) such as 1195 A 15 and C 85 A 10 from Elastollan® TPU. The damping units **50** in the examples shown in FIGS. **13A-13E**, are 3D printed from TPU **90A**. A tested version of the 3D printed damping unit **50** resisted compressive load of 5000 N with 5 millimeters displacement making it suitable for great landing impacts, for skaters weighing around 70 Kilograms. Damping units **50** of different stiffnesses are intended for skaters of different weight categories, generally limited by the displacement of the damping unit **50** being less than one millimeter under a load of two times the body weight of a skater. A tested version of the 3D printed damping unit **50** displaced 1 millimeter under compressive load of 1500 N, making it a suitable version of the damping unit **50** for skaters weighing around 70 Kilograms. A stiffer version of the damping unit **50** can be placed in the front section of the frame **40** and softer in the rear section of the frame **40**.

(57) —Bumpers—

(58) The frame **40** of the modular damping ice-skating blade assembly **90**, in configuration for 2DOF or 1DOF damping, may include bumpers **10** extending downwards from the top inside wall of the frame **40**, in line with the blade runner **70**, protecting the frame **40** from possible impacts of the blade runner **70** during its movement inside the frame **40**. The bumpers **10** may also be used to limit the displacement of the damping unit **50** by stopping the blade runner **70** at a specific distance from the frame **40**. The bumpers **10** may also be used as additional cushioning for the blade runner **70**, to improve the impact damping. The bumpers **10**, according to the first embodiment best shown in FIGS. **1-4**, **6** and **7**, are placed in the frames **20** and **60** on locations where possible impacts of the blade runner **70** to the frame **40** may occur, generally where the top portion of the blade runner **70** has the closest vertical distance to the top inside wall of the frame **40**. The bumpers **10**, in this example, include protrusions **11** extending upwards from the top surface of the bumpers **10** configured to fit and hold in grooves **29** located in the top inside wall of the frames **20** and **60**. In an alternative embodiment, the bumpers **10** may be glued to the frame **40** or may be an integral part of the damping unit **50**.

(59) The invention is not limited to the embodiments but can be varied within the scope of the appended claims.

Claims

1. A configurable modular damping ice-skating blade assembly comprising: a frame comprising a plurality of protruding arms; a blade runner removably mounted in relation to the frame and extending along a longitudinal axis, the blade runner being one of a plurality of modified blade runners configured to be received in the frame; and at least one damp device disposed in the frame

to dampen a movement of the frame in relation to the blade runner, the at least one damping device being located between the protruding arms of the frame, the protruding arms being configured to secure the damping device in the longitudinal axis relative to the frame, wherein the configurable modular damping ice-skating blade assembly is configurable with each of the plurality of modified blade runners to damp movement of the frame in relation to the blade runner.

2. The assembly according to claim 1, wherein the at least one damping device is disposed in said frame with a configuration to allow said blade runner to move in a flexible manner in said frame along the longitudinal axis and along a vertical axis which is perpendicular to said longitudinal axis, to provide damping in two degrees of freedom.

3. The assembly according to claim 2, wherein said blade runner is rigidly mounted in relation to said frame creating a rigid configuration substantially without damping.

4. The assembly according to claim 2, wherein said blade runner is provided with at least one large transversally extending aperture allowing the blade runner to be attached to said frame, or said blade runner is provided with at least one large transversally extending aperture and one generally circular through-hole allowing the blade runner to be attached to said frame, said through-hole being configured to pivotally couple with the frame, or said blade runner is provided with at least two generally circular through-holes allowing the blade runner to be attached to said frame creating a rigid construction.

5. The assembly according to claim 4, wherein the blade runner is attached to the frame by a transversal bearer extending through said large transversally extending aperture or said circular through-hole.

6. The assembly according to claim 5, wherein the large transversally extending aperture allows movement of the blade runner upwards to different angles in the forward-rearward longitudinal axis, the vertical axis and along an axis of rotation relative to the frame without interfering with said transversal bearer that passes through the large transversally extending aperture.

7. The assembly according to claim 5, wherein said transversal bearer is configured as a pin with a screw hole and a screw located in said screw hole, to create a counterforce mechanism enabling to adjust the fit and preloading force between the blade runner and the at least one damping device by screwing said screw from said pin against said blade runner which shifts said blade runner upwards in the direction of said at least one damping device.

8. The assembly according to claim 1, wherein said blade runner is pivotally coupled in relation to said frame in a configuration to move in a flexible manner in said frame along an axis of rotation, which is transverse to said longitudinal axis, so as to provide damping in one degree of freedom.

9. The assembly according to claim 1, wherein said at least one damping device has a curved longitudinal profile that enables the blade runner to be secured within the frame in the forward-rearward longitudinal axis and vertical axis relative to the frame.

10. The assembly according to claim 1, wherein the frame comprises a front-frame portion and a back-frame portion, and wherein a first damping device of the at least one damping device is located between protruding arms in the front-frame portion and a second damping device of the at least one damping device is located between protruding arms in the back-frame portion.

11. The assembly according to claim 1, wherein said at least one damping device is provided with a protrusion extending from a top portion thereof to secure a position of the at least one damping device within the frame, said protrusion being configured to fit in an aperture in said frame.

12. The assembly according to claim 1, wherein said at least one damping device is configured as an elastic body with a cavity.

13. The assembly according to claim 12, wherein said at least one damping device comprises an infill or a spring which is accommodated within said cavity in the elastic body, to reinforce the at least one damping device.

14. The assembly according to claim 1, wherein the frame comprises a plurality of bumpers extending downwards from a top inside wall of said frame and in line with the blade runner, the

plurality of bumpers being configured to protect the frame from impacts of the blade runner during movement of the blade runner inside the frame.

15. The assembly according to claim 1, wherein said frame constitutes an integral part of a sole of an ice-skating boot.

16. The assembly according to claim 1, wherein the at least one damping device is one of a plurality of damping devices of different stiffnesses for skaters of different weight categories, limited by displacement of the at least one damping device being less than one millimeter under a load of two times the body weight of a skater.

17. The assembly according to claim 1, wherein the protruding arms of the frame are joined to form an opening configured to accommodate the at least one damping device.

18. The assembly according to claim 1, wherein the protruding arms of the frame are configured to allow lateral displacements of the at least one damping device when a compressive load is transmitted to the at least one damping device.

19. A configurable modular damping ice-skating blade assembly comprising: a frame comprising a plurality of protruding arms; a blade runner being removably mounted in relation to the frame and extending along a forward-rearward longitudinal axis, the blade runner being one of a plurality of blade runners configured to be received in the frame, the blade runner being configured to attach to the frame by a transversal bearer configured to pass through an aperture in the blade runner; and at least one damping device disposed in said frame to allow one or more configurations to damp the frame in relation to the blade runner, the at least one damping device located between the protruding arms of the frame, the protruding arms being configured to secure the at least one damping device in the forward-rearward longitudinal axis relative to the frame, wherein the transversal bearer is configured to enable adjustment of a fit and preloading force between the blade runner and the at least one damping device.

20. A configurable modular ice-skating blade assembly comprising: a frame comprising a plurality of protruding arms; a blade runner extending along a forward-rearward longitudinal axis, the blade runner being configured to attach to the frame, the blade runner being one of a plurality of blade runners configured to be attached to the frame; and at least one resilient system disposed in said frame allow one or more configurations to mitigate load, shocks, and vibrations transmitted to the frame in relation to the blade runner, the at least one resilient system being disposed between the protruding arms of the frame, the protruding arms being configured to secure the at least one resilient system in the forward-rearward longitudinal axis relative to the frame.
